

Adding Two-Digit Numbers with Place Value

Answer Key

Grade 2–3 Arithmetic

Quick Answers

1. 38	3. 65	5. 63	7. 75
2. 59	4. 84	6. 90	8. 73

Common wrong answers

6. *If they got 92: added the exact numbers instead of rounding each one first*
6. *If they got 80: rounded 58 down to the ten below instead of to the nearest ten*
8. *If they got 613: added the tens and the ones as if they were separate little sums and wrote the two results side by side*
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How to use this key. Each solution is written the way a second or third grader should say it out loud: name the tens, name the ones, put them back together. If a child gets the right total by counting on by ones, the answer is correct but the strategy is not yet there — ask them to show it again with the blocks or the number line.

- 1.** Base-ten blocks: a group of 23 (two tens, three ones) plus a group of 15 (one ten, five ones).

Solution: Count the tall bars first: two tens and one ten make three tens, which is 30. Now the small squares: three ones and five ones make eight ones. Three tens and eight ones is 38. Written as a number sentence, $23 + 15 = 38$. The total is 38.

- 2.** Number line: start at 34 and add 25.

Solution: Break 25 into its tens and its ones: 25 is two tens and five ones. Jump the tens first — from 34 to 44, then from 44 to 54. Now jump the five ones: 54 to 59. Landing on 59 means $34 + 25 = 59$, so the total is 59. Jumping the tens first is what keeps the count short: two big jumps and one small one, instead of twenty-five little steps.

- 3.** Base-ten blocks with a trade: 38 (three tens, eight ones) plus 27 (two tens, seven ones).

Solution: Tens first: three tens and two tens make five tens, or 50. Ones next: eight ones and seven ones make fifteen ones. Fifteen ones is more than a full ten, so circle ten of them and trade: that is one more ten and five ones left over. Now there are six tens and five ones. Putting it together, $50 + 15 = 65$, so $38 + 27 =$ 65.

- 4.** Number line: start at 46 and add 38.

Solution: 38 is three tens and eight ones. Jump the three tens: 46 to 56, 56 to 66, 66 to 76. Then jump eight ones. Going from 76 to 84 crosses a ten, so it helps to split the eight: four ones gets to 80, and four more gets to 84. The landing number is the total: $46 + 38 =$ 84.

5. Friendly ten: Rosa adds 39 and 24 by sliding 1 across.

Solution: 39 is only one away from 40, and 40 is easy to add to. So Rosa moves 1 out of the 24: the 39 becomes 40 and the 24 becomes 23. Nothing was created or lost — the same total is just packed differently. Now the addition is friendly: $40 + 23 = 63$. Checking the other way, $39 + 24 = 63$ as well, so the total is 63. Note that Rosa does not have to give anything back at the end: she moved the 1 from one addend to the other rather than adding it on top.

6. Estimate: round 58 and 34 to the nearest ten, then add.

Solution: On the number line 58 sits past the middle of the way from 50 to 60, so it is closer to 60 — round it up. For 34, the 4 ones are less than half of a ten, so 34 rounds down to 30. Adding the rounded numbers, $60 + 30 = 90$, so the sum is about 90. The exact answer is 92, which is close to the estimate — that closeness is how an estimate checks an answer.

7. Friendly ten on the second number: 47 plus 28.

Solution: 28 is two away from 30, so give it 2 and add the easier problem: $47 + 30 = 77$. But this time the 2 was borrowed from nowhere — it was added on top of the real total, so it has to be taken back off: $77 - 2 = 75$. Hence $47 + 28 = \text{75}$. Compare this with problem 5: there the 1 was moved *between* the two numbers, so nothing was given back; here the 2 was added *in*, so it must be removed.

8. Find the mistake: a student wrote $46 + 27 = 613$.

Solution: The student added the tens as if they were single digits ($4 + 2 = 6$) and the ones separately ($6 + 7 = 13$), then wrote the two results next to each other as “613”. The 6 was really six tens, that is 600 in the way it was written, and the two pieces were never actually added together. Doing it with place value instead: six tens and seven ones is thirteen ones, so trade ten of them for a ten. That gives $40 + 20 + 10 = 70$ and three ones left over. So $46 + 27 = \text{73}$. A quick estimate would have caught it: 46 and 27 are close to 50 and 30, so the answer has to be near 80, nowhere near 613.